

LECTURE 4

DATA ANALYSIS

EXPLORING DATA



LECTURE CONTENT



- What is Data Analysis?
- Why Data Analysis?
- Data Analysis Tools
- Types of Data Analysis: Techniques and Methods
- Data Analysis Process
- Excel/Google Sheet features
- Tools for Data Analyst

DATA ANALYSIS

Data analysis is a wide subject and contains many ideas. Data analysis may refer to learning or extracting information from some dataset and use it to analyze the current behavior of phenomena and maybe predict future or past behavior.



DATA ANALYSIS

A simple example of Data analysis is whenever we take any decision in our day-to-day life is by thinking about what happened last time or what will happen by choosing that particular decision. This is nothing but analyzing our past or future and making decisions based on it. For that, we gather memories of our past or dreams of our future. So that is nothing but data analysis. Now same thing analyst does for business purposes, is called Data Analysis.

DATA ANALYSIS TOOLS



SQL



MATLAB

Types of Data Analysis: Techniques and Methods

- Text Analysis
- Statistical Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis



TYPES OF DATA ANALYSIS

TEXT ANALYSIS

Text Analysis is also referred to as Data Mining. It is a method to discover a pattern in large data sets using databases or data mining tools.

STATISTICAL ANALYSIS

Statistical Analysis shows "What happen?" by using past data in the form of dashboards.

DIAGNOSTIC ANALYSIS

Analyses complete data or a sample of summarized numerical data

PREDICTIVE & PRESCRIPTIVE ANALYSIS

Predictive Analysis shows "what is likely to happen" by using previous data. Prescriptive Analysis combines the insight from all previous Analysis to determine which action to take in a current problem or decision.

DATA ANALYSIS PROCESS

- Data Requirement Gathering
- Data Collection
- Data Cleaning
- Data Analysis
- Data Interpretation
- Data Visualization



DATA ANALYSIS PROCESS EXAMPLES

- **Data Requirement Gathering:** Ask yourself why you're doing this analysis, what type of data analysis you want to use, and what data you are planning on analyzing.
- **Data Collection:** Guided by the requirements you've identified, it's time to collect the data from your sources. Sources include case studies, surveys, interviews, questionnaires, direct observation, and focus groups.
- **Data Cleaning:** Not all of the data you collect will be useful, so it's time to clean it up. This process is where you remove white spaces, duplicate records, and basic errors. Data cleaning is mandatory before sending the information on for analysis.

DATA ANALYSIS PROCESS EXAMPLES

- **Data Analysis:** Here is where you use data analysis software and other tools to help you interpret and understand the data and arrive at conclusions. Data analysis tools include Excel, Python, R, Rapid Miner, Chartio, Metabase, Microsoft Power BI and other.
- **Data Interpretation:** Now that you have your results, you need to interpret them and come up with the best courses of action, based on your findings.
- **Data Visualization:** Data visualization is a way of saying, “graphically show your information in a way that people can read and understand it.” You can use charts, graphs, maps, bullet points, or a host of other methods. Visualization helps you derive valuable insights by helping you compare datasets and observe relationships.



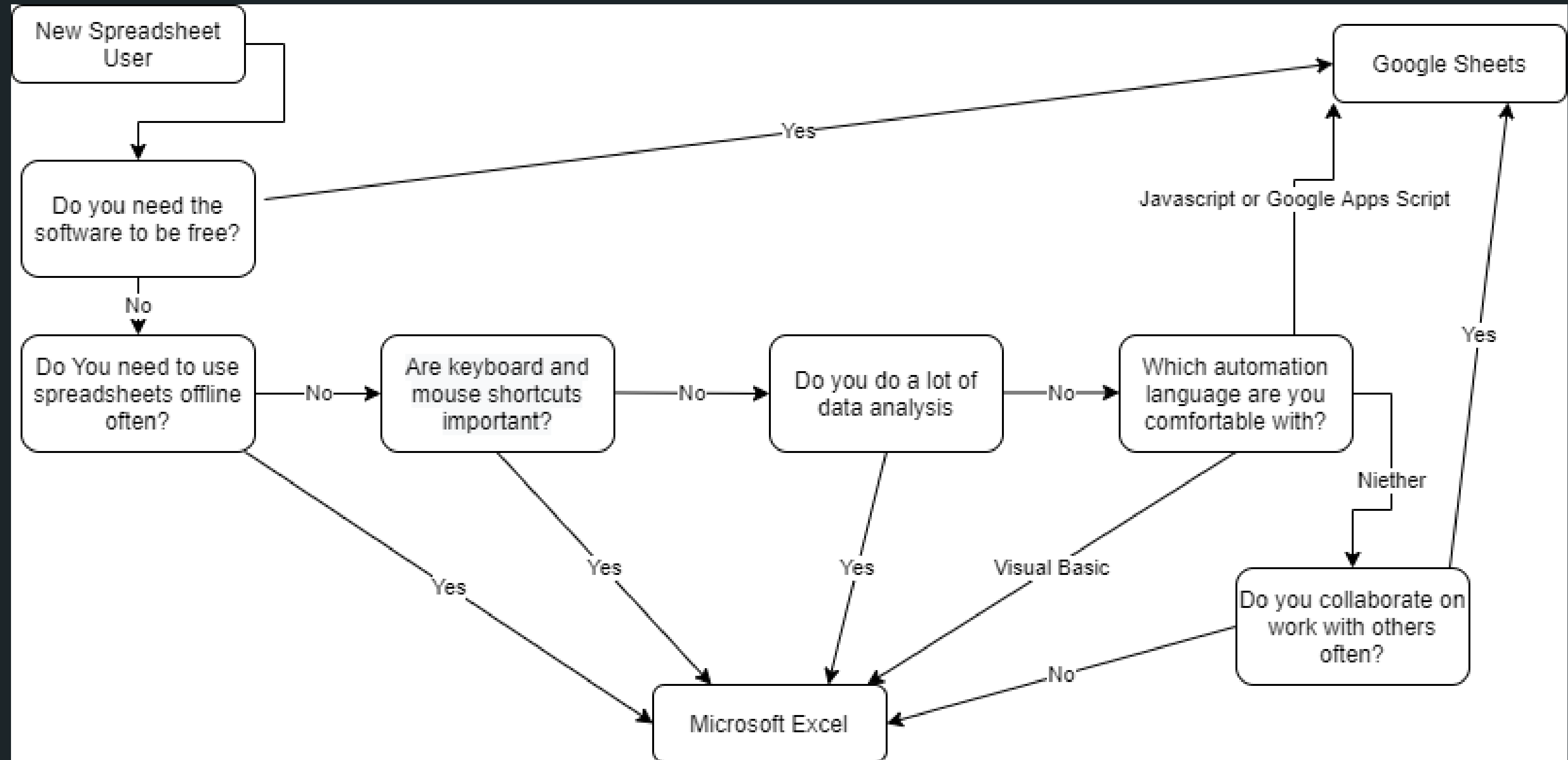
EXCEL & GOOGLE SHEET

Microsoft Excel is one of the most popular tools used in today's fast paced business environment. It is used in almost every job role, across a huge range of industries including manufacturing, finance, logistics, hospitality, healthcare and engineering. It enables you to work with numbers in rows and columns, perform calculations, analyze data and produce superb charts.

Google Sheets is a free online spreadsheet program provided by Google and is part of their G Suite software applications. All you need to access the power of Google sheets is a google account and a browser.

GOOGLE SHEETS VS MICROSOFT EXCEL

FLOWCHART



EXCEL/GOOGLE SHEET'S COMMON DATA ANALYSIS FEATURES

SORT

Sort your Excel/Google Sheet data on one column or multiple columns. It is possible to sort in ascending or descending order.

FILTER

Filter your Excel/Google Sheet data if you only want to display records that meet certain criteria.

CONDITIONAL FORMATTING

Conditional formatting in Excel/Google Sheet enables you to highlight cells with a certain color, depending on the cell's value.

EXCEL/GOOGLE SHEET'S COMMON DATA ANALYSIS FEATURES

CHARTS

A simple Excel/Google Sheet chart can say more than a sheet full of numbers.

PIVOT TABLES

VUI performs actions necessary for the performance of the user's tasks, which the user did not ask.

WHAT-IF ANALYSIS

What-If Analysis in Excel allows you to try out different values (scenarios) for formulas.

SOLVER

Excel includes a tool called solver that uses techniques from operations research to find optimal solutions for all kinds of decision problems.

TOOLS YOU MUST LEARN AS A DATA ANALYST



Excel

A classic in the world of data analysis, Excel tops the list as a crucial tool to learn as a data analyst. It is a straightforward programme to learn, and data analysts should be proficient in all aspects of Excel from using formulas to creating pivot tables. Any spreadsheet application will be useful, but Microsoft Excel is the most widely used one.



Many large companies use SQL for data analysis, and it is considered to be one of the key tools for analysts. It is one of the most prominent tools you must learn as a data analyst. SQL is also used by software engineers in software development. SQL is a programming language that was initially created for managing data from a relational database. It is a straightforward tool to learn and can be used for complex, challenging data analytics. Additionally, it lets you collate data in a similar manner to Excel but over large datasets and across multiple tables at once.

TOOLS YOU MUST LEARN AS A DATA ANALYST



Frequently used in inferential statistics and probability, Python is an open-source programming language that lets you build and manage data structures with the Pandas library. Python is a versatile tool that supports data manipulation, data analysis and data representation. Many companies that have an online presence used Python to analyse user activity, for example. Python is especially good for creating graphics and visual representations of numerical information. It lets you visualise data collected by presenting it in charts or graphs, and it can also be used to interpret large datasets. Python is also a useful tool if you need to test a hypothesis through A/B testing and to evaluate the data collected.

TOOLS YOU MUST LEARN AS A DATA ANALYST



Data Management tools - R, SAS and SPSS

Three hugely important tools in data analysis, in particular data management, are R, SAS and SPSS. Data management is the process of organising, storing and maintaining data within an organisation or company. It constitutes an important step in data analytics, as the information may need to be instantly available, up to date and accessible for various analytical functions.

R

Compatible with all platforms - Windows, Mac OS and Linux, the open-source programming language R is another essential and commonly used tool in data analytics. It is broadly used by statisticians for statistical modeling, as it offers a great range of statistical and graphical possibilities, and it is commonly used to perform data wrangling. It lets you produce visuals such as plots and graphs to represent data and is available across various libraries such as Plotly. It's used in finance, sales but also some scientific fields such as medicine and technology.

TOOLS YOU MUST LEARN AS A DATA ANALYST



Power BI and Tableau are two important Business Intelligence (BI) technologies for the collection, integration, analysis, and presentation of business information. They help you perform data analysis, data manipulation, and data visualization to make sense of business data and draw insights.

